RAG TEST KNOWLEDGE BASE

Sample Reference Document for Retrieval-Augmented Generation Testing

How to use this document

This document is intentionally designed for testing a Retrieval-Augmented Generation (RAG) application. It contains multiple topics, specific facts, definitions, comparisons, and named examples. The information is fictional or educational and can be queried after the document is uploaded to a RAG system.

Topic Index

1. Cloud Computing — service models, deployment models, benefits and limitations.
2. Artificial Intelligence — AI, machine learning, deep learning and applications.
3. Cybersecurity — common threats, security controls and best practices.
4. Data Analytics — data types, analytics methods and a small example dataset.
5. Green Technology — renewable energy, smart buildings and sustainable computing.

Suggested RAG test: Ask questions whose answers appear on different pages. Then test summarization, comparison, multi-hop questions, and questions for which the document contains no answer.

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1. CLOUD COMPUTING

1.1 Definition

Cloud computing is a computing model in which users access computing resources such as servers, storage, databases, networking, and software through a network. Instead of maintaining all infrastructure locally, an organization can obtain resources from a cloud provider and scale them according to demand.

1.2 Service Models

Infrastructure as a Service (IaaS): Provides virtual machines, storage, networks, and other infrastructure resources. The customer generally manages the operating system and applications.
Platform as a Service (PaaS): Provides a managed environment for developing and deploying applications. Developers can focus on application code without managing most underlying infrastructure.
Software as a Service (SaaS): Provides complete software applications through a network. The provider manages the application and infrastructure.

1.3 Deployment Models

A **public cloud** provides resources through a shared provider infrastructure. A **private cloud** is dedicated to a single organization. A **hybrid cloud** combines private and public environments so workloads or data can be distributed between them.

1.4 Benefits and Limitations

Major benefits include rapid provisioning, flexible capacity, reduced upfront infrastructure costs, and access to managed services. Important limitations include dependence on network connectivity, possible service outages, vendor lock-in, and the need for careful security and cost management.

Key Fact

The three commonly discussed cloud service models are **IaaS, PaaS, and SaaS**.

2. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

2.1 Artificial Intelligence

Artificial Intelligence (AI) refers to computer systems designed to perform tasks that normally require capabilities associated with human intelligence, such as language understanding, pattern recognition, decision support, and perception.

2.2 Machine Learning

Machine Learning (ML) is a branch of AI in which algorithms learn patterns from data and use those patterns to make predictions or decisions. Common learning categories include supervised learning, unsupervised learning, and reinforcement learning.

2.3 Deep Learning

Deep learning uses neural networks with multiple processing layers. It is widely used for image classification, speech recognition, natural-language processing, and other tasks involving large or complex datasets.

2.4 Example: Student Performance Prediction

Consider a fictional system called **EduPredict**. It predicts whether a student may need academic support using attendance percentage, assignment completion, previous examination marks, and weekly study hours. In one example, a student has 82% attendance, 90% assignment completion, 74 average marks, and 12 weekly study hours. The system classifies the student as **Low Support Risk**.

2.5 AI vs Machine Learning

Concept	Meaning	Example
AI	Broad field of intelligent computer systems	Virtual assistant
ML	Learning patterns from data	Spam detection
Deep Learning	Multi-layer neural network approach	Image recognition

3. CYBERSECURITY, DATA ANALYTICS AND GREEN TECHNOLOGY

3.1 Cybersecurity

Cybersecurity protects computer systems, networks, applications, and data from unauthorized access, disruption, alteration, or destruction. Common threats include phishing, malware, ransomware, password attacks, and social engineering.

3.2 Security Controls

Useful controls include multi-factor authentication, strong passwords, regular software updates, encrypted communication, backups, access control, and employee security awareness training.

3.3 Data Analytics

Data analytics is the process of examining data to discover patterns and support decisions. **Descriptive analytics** explains what happened, **diagnostic analytics** investigates why it happened, **predictive analytics** estimates what may happen, and **prescriptive analytics** suggests possible actions.

3.4 Example Dataset

A fictional online store records four products. Product A has 120 units sold at ■450 per unit. Product B has 85 units sold at ■700 per unit. Product C has 150 units sold at ■300 per unit. Product D has 60 units sold at ■900 per unit. Based on units sold alone, Product C has the highest sales volume.

3.5 Green Technology

Green technology focuses on reducing environmental impact through efficient resource use and cleaner technologies. Examples include solar power, energy-efficient buildings, electric mobility, efficient data centers, battery storage, and smart energy-management systems.

RAG Challenge Questions

1. What are the three cloud service models?
2. What attendance does the EduPredict example student have?
3. Which product has the highest sales volume?
4. What is the difference between predictive and prescriptive analytics?
5. Name two cybersecurity controls mentioned in the document.
6. What is the fictional system used in the AI example?
7. Does this document mention quantum computing? If not, say that the answer is not present.

End of RAG Test Document — RAG-DEMO-001